

<page_start no = 1>

<sol_start id=1>

$$\begin{aligned} 1. \quad & 16 + 8 - (7p + 3) \\ & 24 - 7p - 3 \\ & 21 - 7p = 0 \\ & -7p = -21 \\ & P = 3 \\ & 1) \ 3 \end{aligned}$$

<sol_end>

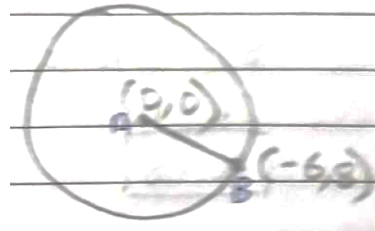
<sol_start id=2>

$$2. \quad 4) \text{ No solutions}$$

<sol_end>

<sol_start id=3>

$$\begin{aligned} 3. \quad & \text{radius} = AB \\ & = \sqrt{(+6)^2 + (-8)^2} \\ & = \sqrt{36 + 64} \\ & = \sqrt{100} \\ & \text{Radius} = 10 \text{ units} \\ & 1) \ 10 \text{ units} \end{aligned}$$



<sol_end>

<sol_start id=4>

$$4. \quad 2) \text{ Proportional}$$

<sol_end>

<sol_start id=5>

$$\begin{aligned} 5. \quad & \frac{\sqrt{3} \sin \theta}{\cos \theta} = 2 \sin \theta \\ & \frac{\tan \theta}{\sin \theta} = \frac{2}{\sqrt{3}} \\ & \frac{1}{\cos \theta} = \frac{2}{\sqrt{3}} \end{aligned}$$

$$\cos \theta = \frac{\sqrt{3}}{2} = \cos 30$$

$$\sin^2 30 - \cos^2 30$$

$$\frac{1-3}{4} = \frac{-2}{4} = \frac{-1}{2}$$

$$2) \frac{-1}{2}$$

<sol_end>

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<page_start no = 2>

<sol_start id=6>

6. 1) $x^3 y^3$

<sol_end>

<sol_start id=7>

7. $\alpha + \beta = \frac{-b}{a} = \frac{2}{4} = \frac{1}{2}$

$$\alpha \beta = \frac{c}{a} = \frac{k-4}{4}$$

$$A/a$$

$$\frac{1}{2} = \frac{k-4}{4} \times \frac{1}{2}$$

$$\frac{1}{2} = \frac{k-4}{8}$$

$$8 = 2k - 8$$

$$2k = 16$$

$$k = 8$$

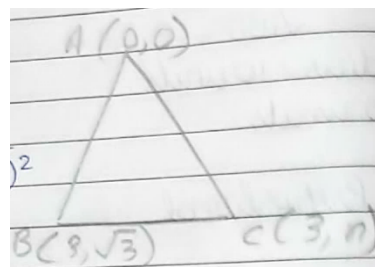
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<sol_start id=8>

8. as ABC is equilateral,

$$AB = AC$$

$$\sqrt{(-3)^2 + (-\sqrt{3})^2} = \sqrt{(-3)^2 + (-x)^2}$$



$$x + 3 = x + x$$

$$x = 3$$

<sol_end>

<sol_start id=9>

9. For equation to have one solution,

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\text{In } \begin{aligned} 2px + 3y &= 7, \\ 2x + y &= 6 \end{aligned}$$

$$a_1 = 2p, a_2 = 2,$$

$$b_1 = 3, b_2 = 1$$

$$\frac{2p}{2} \neq \frac{3}{1}$$

$$p \neq 3$$

P should be 3 but can be any other number.

<sol_end>

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<page_start no = 3>

<sol_start id=10>

10. Given

$$PD \parallel BC,$$

$$\frac{AD}{DC} = \frac{CE}{BE}$$

To prove,

$$PE \parallel AC$$

Proof,

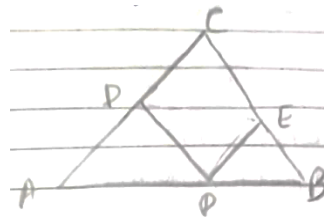
$$\text{As } AD \parallel BC$$

$$\frac{AD}{DC} = \frac{AP}{PB} \quad [\text{BPT}] \quad \dots\dots(i)$$

Also,

$$\frac{AD}{DC} = \frac{CE}{BE} \quad (\text{given}) \quad \dots\dots(ii)$$

From equation (i) and (ii)



$$\frac{AP}{PB} = \frac{CE}{BE}$$

By converse of BPT,

$PE \parallel AC$

Hence, proved.

<sol_end>

<sol_start id=11>

$$11. \quad \tan \theta = \frac{1}{2} = \frac{P}{B}$$

$$H = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\sin \theta = \frac{P}{H} = \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{B}{H} = \frac{2}{\sqrt{5}}$$

$$\text{So, } \frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$$

$$\frac{1}{2} \frac{(1)}{\sqrt{5}} \frac{(2)}{\sqrt{5}} = \frac{1}{5}$$

$$\frac{1}{5} = \frac{1}{5}$$

LHS=RHS

Hence proved

<sol_end>

<page_end>

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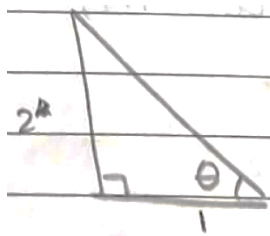
<sol_start id=12>

$$12. \quad \frac{257}{5000} = \frac{257}{2^3 \times 5^4}$$

It is a terminating decimal.

<sol_end>

<sol_start id=13>



13. To prove,

$\sqrt{3}$ is irrational,

Proof

To the contrary, let's assume

$\sqrt{3}$ is rational, it,

$$\sqrt{3} = \frac{a}{b} \text{ where } a, b \text{ are co-prime}$$

$$3 = \frac{a^2}{b^2} \text{ (squaring both sides)}$$

$$b^2 = \frac{a^2}{3}$$

a^2 is divisible by 3

so a is also divisible by 3,

Now, let $a = 3m$,

$$3 = \frac{9m^2}{b^2}$$

$$b^2 = \frac{3m^2}{1}$$

$$m^2 = \frac{b^2}{3}$$

b^2 is divisible by 3,

b is also divisible by 3

This contradicts the fact that a & b are co-prime.

This contradiction occurred due to our wrong assumption.

So,

$\sqrt{3}$ is irrational,

Hence, proved.

<sol_end>

<sol_start id=14>

14. $p(x) = 2x^2 - 3x + 7$

$$\alpha + \beta = \frac{-b}{a} = -\frac{(-3)}{2} = \frac{3}{2}$$

$$\alpha\beta = \frac{c}{a} = \frac{7}{2}$$

$$\frac{1}{2\alpha-3} + \frac{1}{2\beta-3}$$

$$\frac{(2\beta-3)+(2\alpha-3)}{(2\alpha-3)(2\beta-3)} = \frac{2\beta-3+2\alpha-3}{4\alpha\beta-6\alpha-6\beta+9}$$

$$= \frac{2\alpha+2\beta-6}{4\alpha\beta-6\alpha-6\beta+9}$$

$$= \frac{2(\alpha+\beta)-6}{14-\cancel{6}\left(\frac{3}{2}\right)+9} = \frac{-3}{14}$$

$$\frac{-3}{14}$$

<sol_end>

<sol_start id=15>

15. For it to have infinite solutions,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$3x - (a+1)_y = 2b - 1$$

$$5x + (1-2a)_y = 3b$$

$$a_1 = 3, b_1 = (-a-1), c_1 = -2b + 1$$

$$a_2 = 5, b_2 = (1-2a), c_2 = -3b$$

$$\frac{3}{5} = \frac{-a-1}{1-2a}$$

$$3-6a = -5a-5$$

$$8 = a$$

$$\frac{3}{5} = \frac{1-2b}{-3b}$$

$$-9b = 5-10b$$

$$5 = b$$

<sol_end>

<page_end>

<page_start no = 5>

<sol_start id=16>

16. Given

$$\triangle ABE \cong \triangle ACD$$

To prove

$$\triangle ADE \sim \triangle ABC$$

Proof,

$$\text{As } \triangle ABE \cong \triangle ACD$$

$$\left. \begin{array}{l} AB = AC \\ AD = AE \end{array} \right\} [\text{CPCT}] \dots\dots(i)$$

$$\dots\dots(ii)$$

In $\triangle ADE$ and $\triangle ABC$,

dividing both eqs

$$\frac{AD}{AB} = \frac{AE}{AC}$$

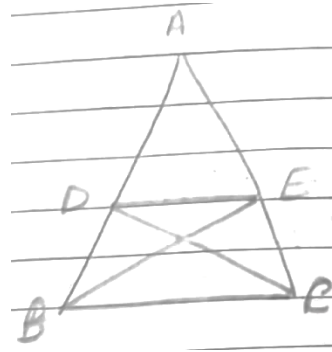
$$\angle A = \angle A \text{ (common)}$$

By SAS

$$\triangle ADE \sim \triangle ABC$$

Hence, proved.

<sol_end>



<sol_start id=17>

$$17. \quad \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta}$$

$$= \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} + \frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\frac{1}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{(\sin^2 \theta - \cos^2 \theta)}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} = \text{RHS}$$

Hence, proved.

<sol_end>

<sol_start id=18>

18. Given

$C(1, 1), A(-2, 7)$

$m_1 = 3, m_2 = 2,$

using section formula,

$$x = \frac{m_1 n_2 + m_2 n_1}{m_1 + m_2}$$

$$1 = \frac{3(x_2) + (2)(-2)}{5}$$

$$5 = 3x_2 - 4$$

$$3x_2 = 9$$

$$\boxed{x_2 = 3}$$

$$y = \frac{m_1 y_2 + m_2 y}{m_1 + m_2}$$

$$1 = \frac{(3)(y_2) + (2)(7)}{5}$$

$$5 = 3y_2 + 14$$

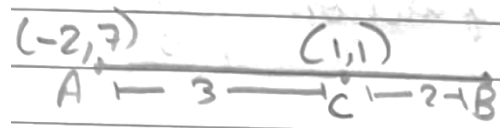
$$3y_2 = -9$$

$$y_2 = -3$$

Coordinates of B $\boxed{3 = -3}$

<sol_end>

<sol_start id=19>



19. let fraction be $\frac{x}{y}$

Given

$$\frac{2x}{y+1} = 1 \Rightarrow 2x = y + 1 \quad \dots\dots(i)$$

$$\frac{x+4}{2y} = \frac{1}{2} \Rightarrow x + 4 = y \quad \dots\dots(ii)$$

Solving both equation

$$2x - y - 1 = 0$$

$$-x + y - 4 = 0$$

$$\boxed{x = 5}$$

$$\boxed{y = 9}$$

$$\text{Fraction} = \frac{5}{9}$$

<sol_end>

<page_end>

<page_start no = 6>

<sol_start id=20>

20. Thales theorem - If a line is drawn parallel to one side of triangle and intersects other two sides at distinct points then other two sides are divided in same ratio.

Given,

ABC is a triangle

DE||BC,

To prove,

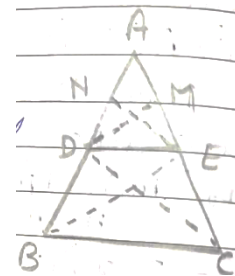
$$\frac{AD}{DB} = \frac{AE}{EC}$$

Construction

Join BE and CD,

Draw $DM \perp AC$ and $EN \perp AB$,

Proof,



$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\cancel{\frac{1}{2}} \times AD \times \cancel{EN}}{\cancel{\frac{1}{2}} \times BD \times \cancel{EN}}$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{AD}{BD} \quad \dots\dots\dots(i)$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CDE)} = \frac{\cancel{\frac{1}{2}} \times AE \times \cancel{DM}}{\cancel{\frac{1}{2}} \times EC \times \cancel{DM}}$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CDE)} = \frac{AE}{EC} \quad \dots\dots\dots(ii)$$

$$\text{ar}(\triangle BDE) = \text{ar}(\triangle CDE) \quad \dots\dots\dots(iii)$$

[Δ 's on same base, between same parallel lines]

from eq. (i), (ii) and (iii),

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence, proved.

<sol_end>

<sol_start id=21>

$$\begin{aligned} 21. \quad & \frac{\sin 30^\circ + 2\cos^2(45) + \tan^2(60)}{\frac{1}{2}\cot 45 + \cos^2(30) + \tan^2(45)} \\ &= \frac{\frac{1}{2} + 2 \cdot \left(\frac{\sqrt{2}}{2}\right)^2 + \tan^2(60)}{\frac{1}{2}\cot 45 + \cos^2(30) + \tan^2(45)} \\ &= \frac{\frac{1}{2} + 1 + (\sqrt{3})^2}{\frac{1}{2}\cot 45^\circ + \cos^2(30) + \tan^2(45)} \end{aligned}$$

$$= \frac{\frac{9}{2}}{\frac{1}{2} + \frac{3}{4} + \tan^2(45)}$$

$$= 2$$

<sol_end>

<page_end>

<page_start no = 7>

<sol_start id=22>

22. Join AC and BD, O is midpoint of AC & BD

So,

Midpoint of AC

$$\rightarrow \left(\frac{6+1}{2}, \frac{5+2}{2} \right) = \left(\frac{7}{2}, \frac{7}{2} \right)$$

Midpoint of BD

$$\rightarrow \left(\frac{2+a}{2}, \frac{3+b}{2} \right)$$

So as both points are same,

$$\frac{2+a}{2} = \frac{7}{2}$$

$$a = 5$$

$$\frac{3+b}{2} = \frac{7}{2}$$

$$b = 4$$

Coordinates of D = (5, 4)

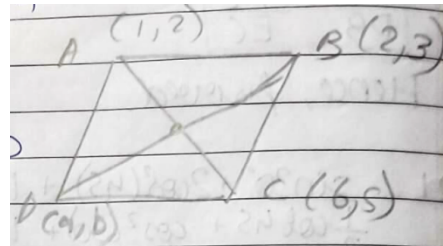
<sol_end>

<sol_start id=23>

23. A will complete 1 round = $\frac{1980}{330} = 6 \text{ min}$

B will complete 1 round = $\frac{1980}{198} = 10 \text{ min}$

C will complete 1 round = $\frac{1980}{220} = 9 \text{ min}$



Let them meet after time t , they've completed n_1 , n_2 & n_3 rounds, all are n dist away from start point

So,

$$330 t = n + 1980 n_1$$

$$198 t = n + 1980 n_2$$

$$220 t = n + 1980 n_3$$

$$\rightarrow 198 t = 330 t - 1980 n_1 + 1980 n_2$$

$$1980(n_1 - n_2) = 132 t$$

$$t = 15(n_1 - n_2)$$

$$220 t = 330 t - 1980 n_1 + 1980 n_3$$

$$1980(n_1 - n_3) = 110 t$$

As t is same and $n_1 - n_3$ or $n_1 - n_2$ can be whole no,

LCM of 15, 18

= 90 min

$\therefore$ They will meet after 90 minutes.

<sol_end>

<sol_start id=24>

24. let,

Son's age = x

Father's age = y

A/a,

$$2x + y = 70 \quad \dots\dots\dots(i)$$

$$2y + x = 92 \quad \dots\dots\dots(ii)$$

Multiplying eq. (1) by 2 and subtracting from eq. (ii)

$$\begin{array}{r} 2y + x = 92 \\ -2y - 4x = -140 \\ \hline -3x = -48 \end{array}$$

$$x = 16$$

Putting in eq. (ii)

$$2y = 80$$

$$y = 40$$

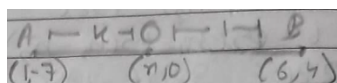
<sol_end>

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<sol_start id=25>

25. Using section formula,



$$y = \frac{m_1 y_2 + m_2 y_1}{x_1 + x_2}$$

$$0 = \frac{(x)(4) + (1)(-7)}{x + 1}$$

$$4x - 7 = 0$$

$$x = \frac{7}{4}$$

$$\text{ratio} = \frac{x}{1}$$

$$= \frac{7}{4} \times \frac{1}{1} = 7:4$$

<sol_end>

<sol_start id=26>

$$26. \quad m^2 n = (\sec \theta - \cos \theta)^2 (\sec \theta - \cos \theta)$$

$$= \left(\frac{1 - \sin^2 \theta}{\sin \theta} \right)^2 \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right)$$

Similarly,

$$mn^2 = (\csc \theta - \sin \theta)^2 (\csc \theta - \sin \theta)$$

$$= \frac{\sin^4 \theta \cos^2 \theta}{\cos^2 \theta \sin \theta} = \sin^3 \theta$$

$$\left(m^2 n \right)^{\frac{2}{3}} + \left(mn^2 \right)^{\frac{2}{3}} = \sin^2 \theta + \cos^2 \theta$$

$$= 1$$

Hence, proved

<sol_end>

<sol_start id=27>

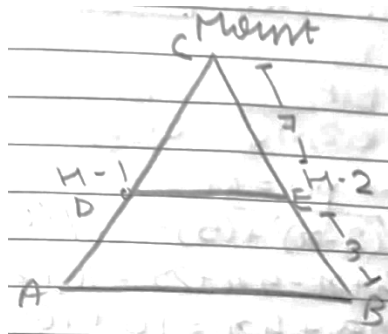
27. Clearly $DE \parallel AB$,

So, In $\triangle ACB$ and $\triangle DCE$,

$\angle C = \angle C$ (common)

$$\frac{AD}{DC} = \frac{BE}{EC}$$

By SAS, $\triangle ACB \sim \triangle DCE$



$$\frac{BC}{EC} = \frac{10}{7} = 10:7$$

<sol_end>

<sol_start id=28>

<sub_id = 1>

28.(i) Using section formula

$$x = \frac{m_1 n_2 + m_2 n_1}{m_1 + m_2}$$

$$x = \frac{(1)(8) + (2)(2)}{3} = \frac{8+4}{3} = \frac{12}{3}$$

$$x = 4$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

$$y = \frac{(1)(3) + (2)(5)}{3} = \frac{3+10}{3} = \frac{13}{3}$$

$$\text{Coordinates of } Q = \left(4, \frac{13}{3}\right)$$

<sub_id end>

<sub_id = 2>

(ii) PR

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-2)^2 + (3-5)^2}$$

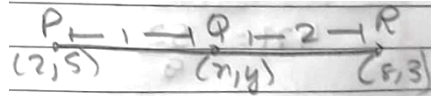
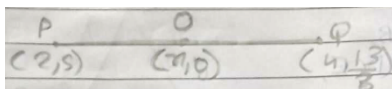
$$= \sqrt{36+4}$$

$$= \sqrt{40} = 2\sqrt{10} \text{ units}$$

<sub_id end>

<sub_id = 3>

(iii)



$$\sqrt{(2-x)^2 + (5)^2} = \sqrt{(4-x)^2 + \left(\frac{13}{2}\right)^2}$$

$$4 + \cancel{x^2} - 4x + 25 = 16 + \cancel{x^2} - 8x + \frac{169}{9}$$

$$29 + 4x = 16 + \frac{169}{9}$$

$$13 + 4x = \frac{169}{9}$$

$$\cancel{52 + 16x + 169}$$

$$16x = 117$$

$$x = 117$$

$$4x = \frac{169 - 117}{9}$$

$$4x = \frac{52}{9}$$

$$x = \frac{\cancel{52} \ 13}{\cancel{98} \ 4}$$

$$x = \frac{13}{4}$$

<sub_id end>

<sub_id = 4>

(iv) Midpoint of PR

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{12}{2}, \frac{8}{2} \right) = (5, 4)$$

<sub_id end>

<sol_end>

<page_end>

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<sol_start id=29>

29. let speed of A = x km/hr

let speed of B = y km/hr

when traveling in same direction,

relative speed $\rightarrow x - y$

dist = 100 km

t = 4 hr

dist = $s \times t$

$$100 = (x - y)4$$

$$x - y = 25 \quad \dots(i)$$

when moving in opp. direction

relative speed $\rightarrow x + y$

dist = 100 km

t = 1 hr

dist = $s \times t$

$$100 = (x + y)$$

$$x + y = 100 \quad \dots(ii)$$

Solving both eq

$$x - y = 25$$

$$x + y = 100$$

$$2x = 125$$

$$x = 62.5 \text{ km / hr}$$

$$y = 37.5 \text{ km / hr}$$

<sol_end>

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